



Visualizing NOAA Climate Data

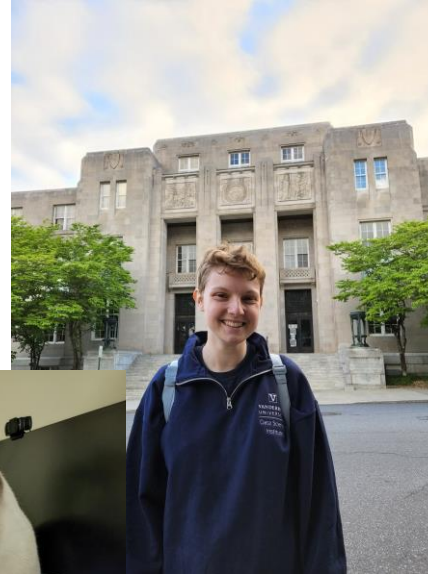
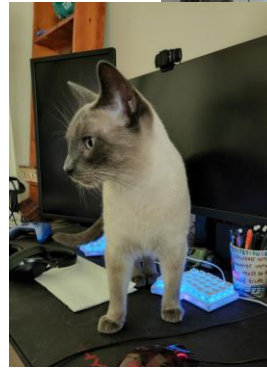
National Centers for
Environmental Information (NCEI)

9/30/2026

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Vanderbilt University

About Me...

- Lecturer / DS at VU College of Connected Computing
- Former Data Scientist at NOAA NCEI, Archive Branch
 - Data engineering
 - Cloud computing
 - Machine learning and AI
- Former NASA Intern



What is NCEI?

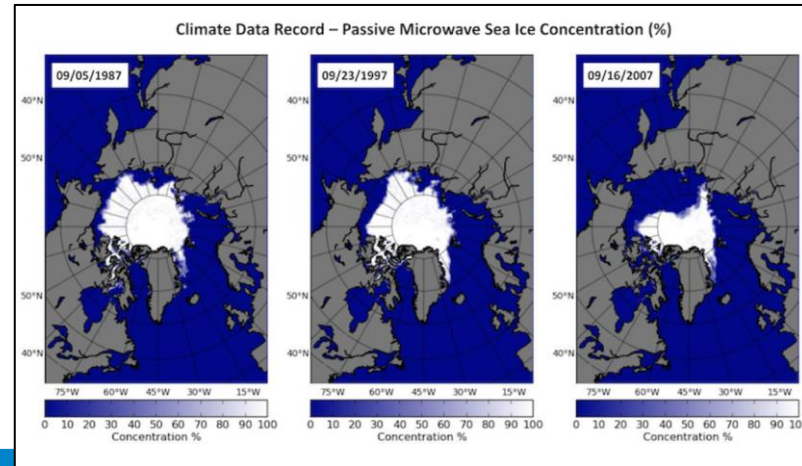
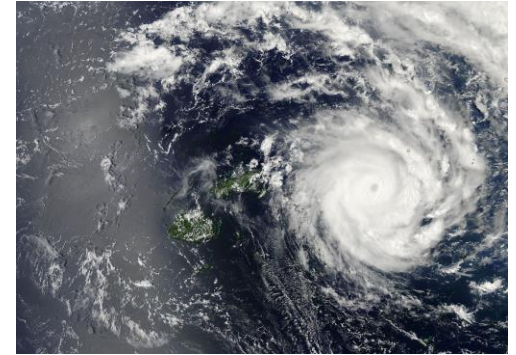
- National Centers for Environmental Information
- Hosts and provides access to one of the most significant archives on Earth
- US leading authority for environmental data

“From the bottom of the ocean to the surface of the sun!”



What can you do with climate data?

- Visualize
 - Ocean/Land surface temperature
 - Ocean currents
 - Natural disasters
- Analyze
 - Human impact
 - Sea ice concentration
 - Vegetation cover
- Make predictions
 - Weather
 - Drought
 - Floods



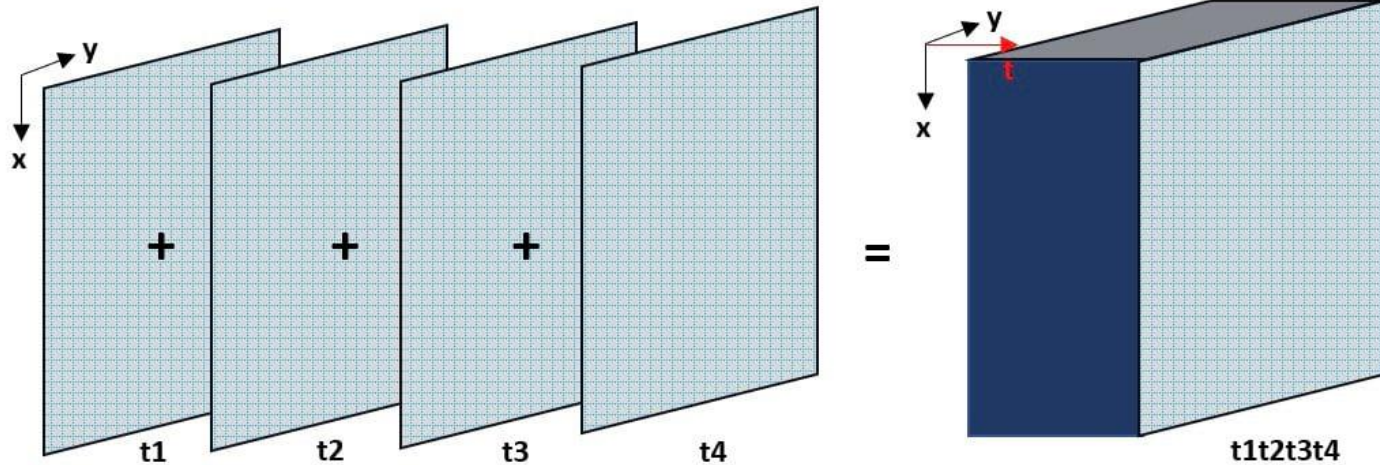
Storing Climate Data

Great for
maps!

- NetCDF: gridded data
 - *.nc*
 - Stores data in a matrix of longitude and latitude
 - Separate matrix for each time step
 - Called “stacks” or “layers”
- NetCDF file contains its own metadata
 - Variable names and definitions, units of measurement, etc.
- Support mathematical operations, such as:
 - Addition, subtraction, scalar multiplication

Storing Climate Data

Netcdf example



Using NetCDF in R

- RNetCDF library
 - Read, write, and create NetCDF files
- Useful functions
 - **open.nc()** - open a NetCDF file
 - **var.get.nc()**- get data from a specific variable
 - **close.nc()**- close the NetCDF file
- NetCDF tools for other languages
 - Python: NetCDF4, xarray, numpy
 - CLI/Bash: CDO

Accessing Climate Data

- Direct Download
 - Visiting data.noaa.gov and downloading data yourself
 - Takes up memory on your computer
- Data Server
 - Data hosted on a server at NOAA
 - Accessible via code
- Cloud Streaming
 - Accessing data from a cloud server (AWS, Google Cloud, Microsoft Azure)
 - Streams data from cloud provider (may need cloud account)



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Common for
Environmental
Scientists

NOAA Data Servers

- THREDDS (Thematic Real-time Environmental Distributed Data Services)
 - Web-facing directory containing dataset collections
 - Accessible via many web protocols (we will use OPENDAP)
 - Allows you to load data directly into your program
 - View NOAA THREDDS catalog at ncei.noaa.gov/thredds/catalog.html



Putting it all together!

The Plan:

1. Load a NetCDF file from THREDDS into R notebook
2. Explore the NetCDF data
3. Plot the data to create a map

Demo: Visualizing Ocean Temperatures in R

[GitHub Repo](#)



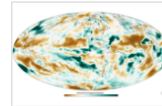
How *Does* Data Shape Our World?

- Climate.gov
 - Taken down in 2025
- Climate.us
 - Built by ex-NOAA employees
 - “successor” to climate.gov

We can only study
what we can access

Maps & Data

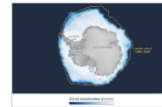
1-10 of 99 results



Precipitation - Global Monthly Percent of Average

Department: Data Snapshots

Was the month drier or wetter than usual? Colors show what percent of its average precipitation a location received in a specific month. White and very light areas had monthly precipitation totals close to the long-term average. Brown areas were drier than average for that month, and blue-green areas were wetter than average for that month.



Antarctic Sea Ice Extent - Maximum

Department: Data Snapshots

How has Antarctic winter sea ice changed from year to year? This collection includes monthly maps for sea ice concentration during the month of the Antarctic winter maximum extent—typically September, rarely October—dating back to the start of the satellite record in 1979. Over that time period, Antarctic sea ice has exhibited an essentially flat trend with substantial year-to-year variability. Beginning in 2016, Antarctic sea ice experienced below-average extents, but as of 2025, determining whether these extents indicate the beginning of an overall downward trend will require at least several more years of observations.

Antarctic Sea Ice Extent - Minimum

Work for NOAA? (or NASA or USGS...etc)

usajobs.gov has many student opportunities





Data Science Institute

Discovery through data.

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